RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject – DevOps and Automation Lab

Lab Assignment 1

Aim:

Install and set up Git & GitHub account.

Learning Objectives

After completing this lab, the student will be able to:

- Understand the purpose of Git and GitHub
- Install Git on a local machine
- Create and configure a GitHub account
- Configure Git with username and email
- Perform basic Git commands such as initialization and commit

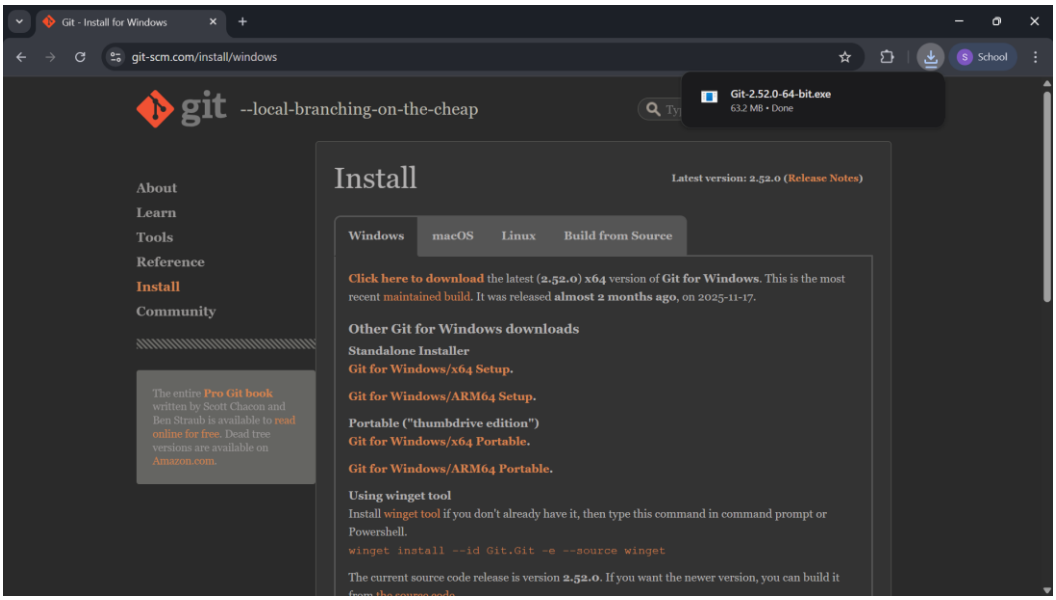
Theory

Git is a distributed version control system used to track changes in source code during software development. GitHub is a cloud-based hosting service that allows developers to store, manage, and collaborate on Git repositories. Together, Git and GitHub enable version control, collaboration, and code management.

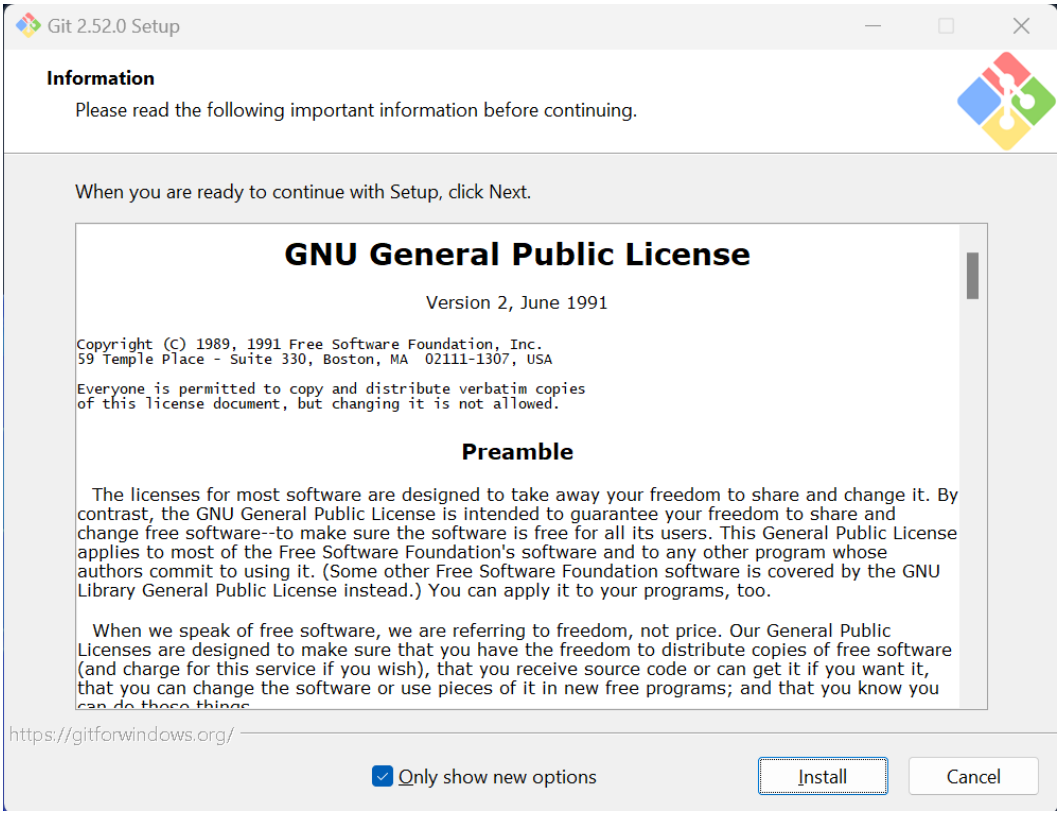
Procedure

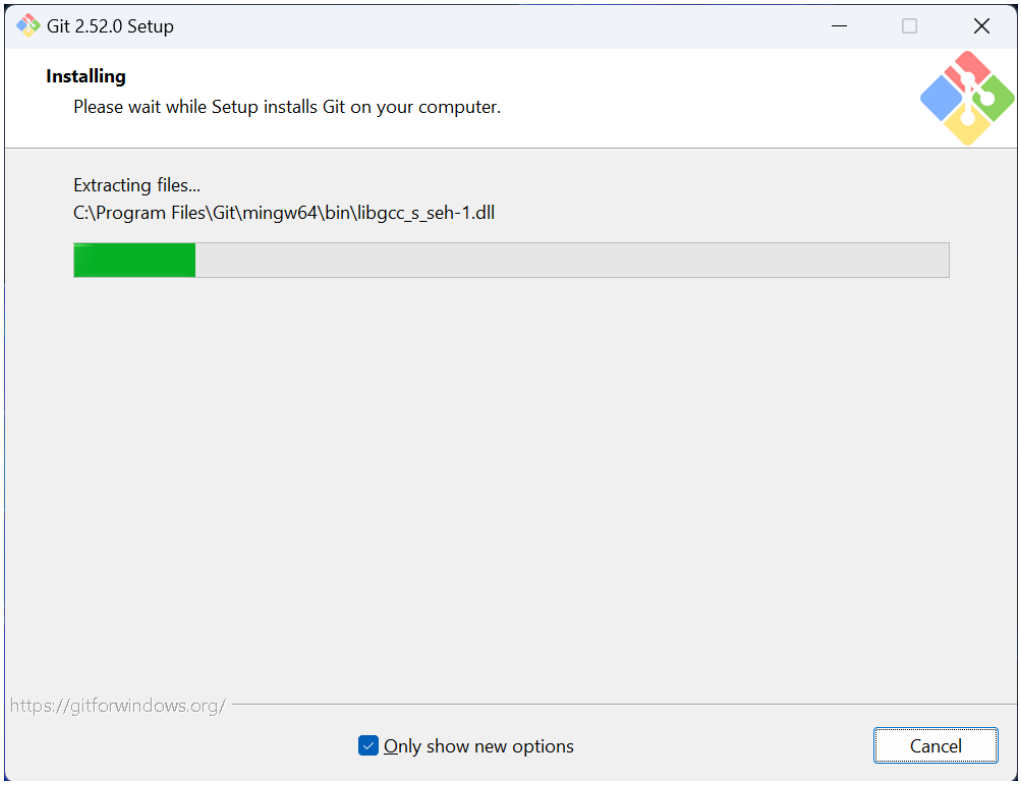
Step 1: Install Git

1. Download Git from the official website.



2. Run the installer and follow the default installation steps.



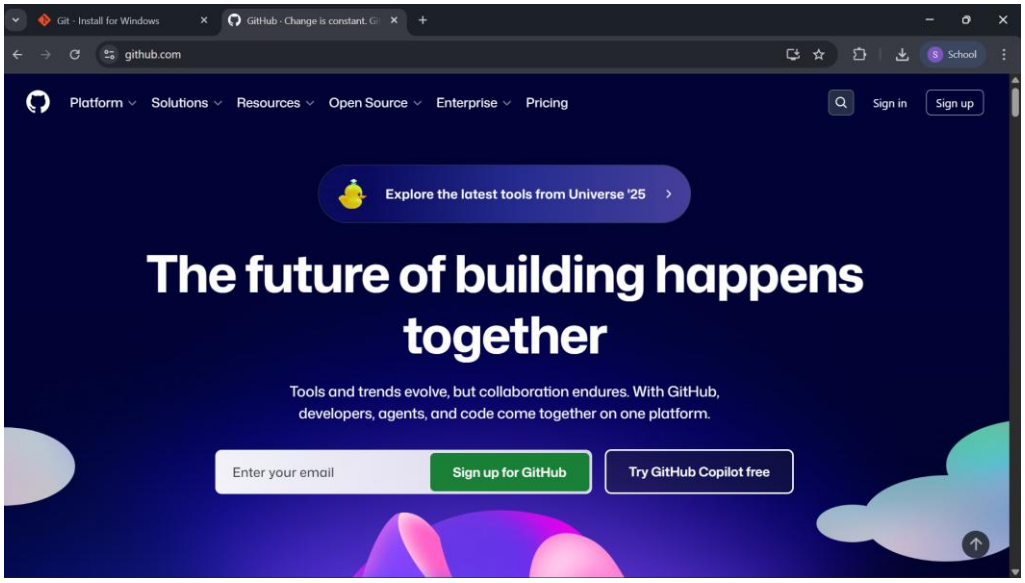


3. Verify installation using the command:
   git –version

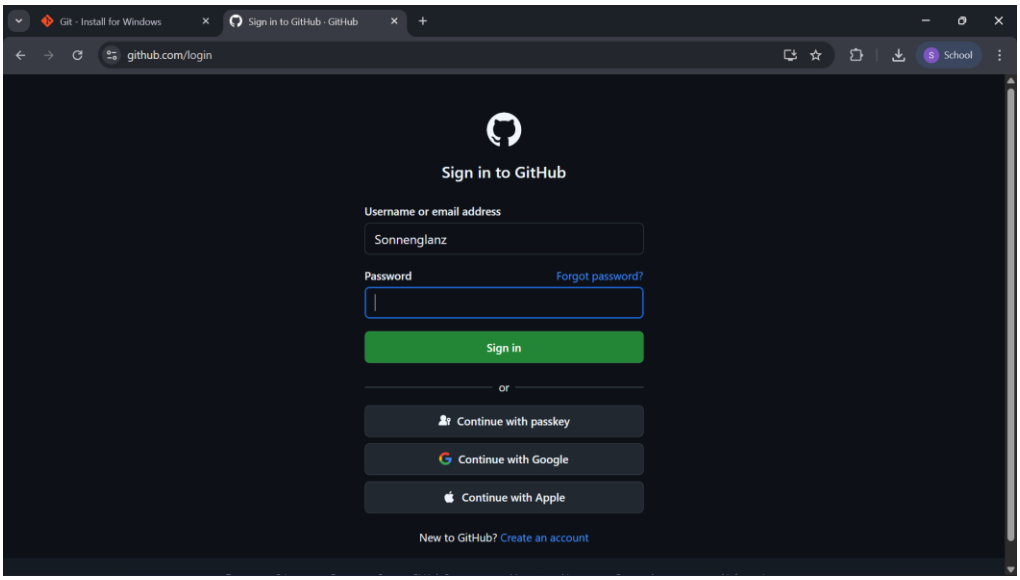
```
shrey@Shreya MINGW64 ~
$ git --version
git version 2.52.0.windows.1
```

Step 2: Create a GitHub Account

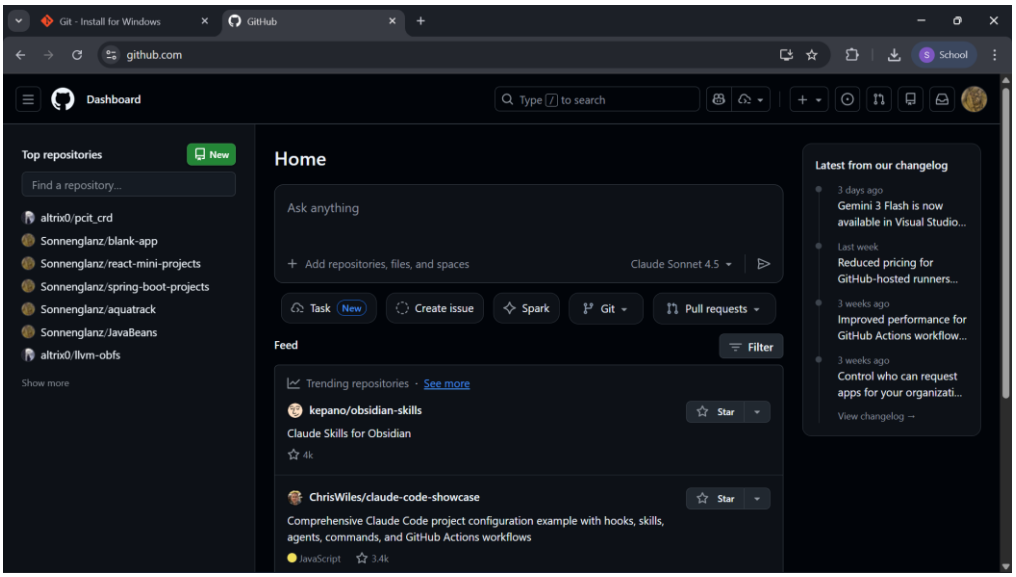
1. Open a web browser and visit GitHub.



2. Sign up using a valid email address.



3. Complete email verification.



Step 3: Configure Git

1. Open Git Bash / Terminal.
2. Set username: git config --global user.name "Your Name"

```
shrey@Shreya MINGW64 ~
$ git config --global user.name "Sonnenglanz"
```

3. Set email: git config --global user.email "your_email@example.com"

```
shrey@Shreya MINGW64 ~
$ git config --global user.email "shreyadpazare@gmail.com"
```

4. Verify configuration: git config –list

```
shrey@Shreya MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
core.editor="C:\Users\shrey\AppData\Local\Programs\Microsoft VS Code\bin\code"
-wait
core.autocrlf=input
user.email=shreyadpazare@gmail.com
user.name=Sonnenglanz
```

Q and A

1. What is Git?

Git is a Distributed Version Control System (DVCS). Unlike older systems that store information as a list of file-based changes (deltas), Git thinks of its data as a series of snapshots of a miniature filesystem. Every time you commit, Git "takes a picture" of what all your files look like at that moment and stores a reference to that snapshot. Key characteristics include:

i. Local Operations: Most operations are nearly instantaneous because they only require local files and resources, rather than information from a server.
ii. Integrity: Everything in Git is checksummed using a SHA-1 hash before it is stored, making it impossible to change file contents or directory structures without Git knowing.
iii. Three States: Git projects have three main sections: the working tree, the staging area (index), and the Git directory (repository).

2. What is the difference between Git and GitHub?

While often used together, they serve different purposes:

Git: Is the actual version control tool that runs locally on your machine to track changes, create branches, and manage snapshots.

GitHub: Is a cloud-based hosting service for Git repositories. It provides a web-based interface, issue tracking, code review tools, and a community platform for developers to collaborate on projects remotely.

3. What is a repository?

A repository (often shortened to "repo") is the directory where Git stores the metadata and object database for your project. This is typically the .git directory that is created when you initialize a project (git init) or cloned when you copy an existing project (git clone). It contains the complete history of the project, allowing you to recall specific versions later.

4. What is a commit?

A commit is the operation of recording the current state of your project from the staging area into the permanent database in your Git directory.

Each commit creates a commit object containing a pointer to the snapshot of the content, metadata like the author's name and email, and a pointer to the commit's parent(s).

It is identified by a unique 40-character SHA-1 hash.

5. What is the use of the git config command?

The git config command is used to get and set configuration variables that control how Git looks and operates. It functions at three levels of priority:

--system: Settings for every user on the system and all their repositories (stored in /etc/gitconfig).

--global: Settings specific to you as a user, affecting all your repositories (stored in ~/.gitconfig). This is used for your primary identity (name and email).

--local: Settings specific to a single repository (stored in .git/config).